



COMPETENCY STANDARD

FOR

JUTE BLEACHING, DYEING, AND PRINTING

(JUTE SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Jute Bleaching, Dyeing, and Printing** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

This document has been developed to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the skills requirements of the Jute sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

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Units & Elements at a Glance:**Generic Competencies (20 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-JUTE-DBP-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-JUTE-DBP-02-G	Work in a team environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hour			20 Hrs.

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-JUTE-DBP-01-S	Work in the Jute Sector	1. Identify the jute sector and its main activities 2. Understand basic roles in jute production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in jute work	10
SICIP-JUTE-DBP-02-S	Apply Green Practices in Jute Sector	1. Interpret green concepts 2. Minimize resources use in the workplace 3. Implement waste management practices	10
Total Hour			20 Hrs.

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Occupation Specific Competencies (180 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-JUTE-JBP-01-O	Interpret Materials and Processes	1. Identify jute fibres and materials 2. Identify dyes and chemical types 3. Identify tools and equipment 4. Describe dyeing and bleaching processes 5. Describe printing process	20
SICIP-JUTE-JBP-02-O	Prepare Materials and Chemicals	1. Prepare jute materials for processing 2. Clean and sort raw materials 3. Mix chemical solutions safely 4. Label and store prepared solutions	20
SICIP-JUTE-JBP-03-O	Perform Bleaching Operations	1. Prepare bleaching chemical solutions 2. Load materials into bleaching machine 3. Carry out bleaching operations 4. Check bleaching results and quality	40
SICIP-JUTE-JBP-04-O	Perform Dyeing Operations	1. Prepare dye solutions and mixtures 2. Load materials into dyeing machine 3. Carry out dyeing operations 4. Check colour quality and consistency	40
SICIP-JUTE-JBP-05-O	Perform Printing Operations	1. Prepare tools, materials, and printing surface 2. Prepare printing paste and colours 3. Apply printing according to design 4. Maintain pattern alignment and consistency	40
SICIP-JUTE-JBP-06-O	Perform Finishing	1. Prepare for the works 2. Identify finishing process	20

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Code	Unit of Competency	Elements of Competency	Duration (hours)
	Operation	3. Carry out finishing 4. Clean and store tools and equipment	
Total Hours			0 Hrs.

COMPETENCY STANDARD: JUTE BLEACHING, DYEING AND PRINTING

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-DBP-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OH&S) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International OHS requirements 1.2 Bangladesh standards for OHS

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	1.3 Building Code 1.4 Fire Safety Rules and Regulations 1.5 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gas Mask 2.3 Gloves 2.4 Safety shoes 2.5 Helmet 2.6 Face mask 2.7 Overalls 2.8 Goggles and safety glasses 2.9 Ear plugs 2.10 Sun block 2.11 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Tourniquet 5.5 Applying CPR (Cardiopulmonary Resuscitation) 5.6 Taking appropriate medicine

Curricular Evidence Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices

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	2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Performing first aid. 2.7 Performing basic firefighting accessories using fire extinguishers 2.8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace. 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness 3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to authority. 1.4 Emergency response plans and procedures are implemented. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-DBP-02-G
Unit Descriptor:		

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This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood and valued. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Team member's support with other members is made to ensure team achieves goals, awareness, and requirements. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of variables:

Variables	Range (may include but not limited to)
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1. Forms of communication	1.1	Agenda
	1.2	Simple reports such as progress and incident reports
	1.3	Job sheets
	1.4	Operational manuals
	1.5	Brochures and promotional material
	1.6	Visual and graphic materials
	1.7	Standards
	1.8	OSH information
	1.9	Signs

Curricular Content Guide

1. Underpinning Knowledge	1.1	Team goals and collaborative decision-making processes
	1.2	Roles and responsibilities of team members
	1.3	Relationships within team and with other workers
	1.4	Effective interpersonal skills to interact with team members
	1.5	Effective formal and informal forms of communication
	1.6	Value of diversity in team functioning.
	1.7	Correct use of workplace terminology
	1.8	Team's duties, responsibilities, authorities, objectives, and task requirements
	1.9	Support mechanism to other members of team to ensure achievements of goals.
	1.10	Methods of identifying current and potential problems faced by a team
	1.11	Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1	Identifying team goals and collaborative decision-making processes
	2.2	Identifying roles and responsibilities of team members
	2.3	Identifying relationships within team and with other workers
	2.4	Using effective interpersonal skills to interact with team members and to contribute to activities and objectives
	2.5	Using formal and informal forms of communication
	2.6	Understanding and valuing views and opinions of other team members
	2.7	Performing tasks in accordance with organizational and team requirements, specifications, and workplace procedures.
	2.8	Supporting other members of the team to ensure team achieves goals, awareness, and requirements.
	2.9	Identifying current and potential problems faced by the team
	2.10	Identifying solutions to the problem
	2.11	Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1	Commitment to occupational health and safety
	3.2	Promptness in carrying out activities
	3.3	Sincere and honest to duties

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	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates, and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 communicated and cooperated with team members. 1.3 worked as a team member 1.4 solved problems as a team member
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE JUTE SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-DBP-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the jute sector. It specifically includes the tasks of identifying the jute sector and its main activities, understanding basic roles in jute production processes, recognizing common tools and equipment used and understanding simple quality requirements in jute work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the jute sector and its main activities	1.1 The jute sector is identified within the context of national and global industries. 1.2 <u>Main activities</u> are recognized. 1.3 <u>Different stages of jute value chain</u> are described in simple terms. 1.4 Key workplaces in the jute sector are identified. 1.5 Importance of the jute sector in the economy and environment is understood.
2. Understand basic roles in jute production processes	2.1 <u>Basic job roles in jute production</u> are identified and described. 2.2 Simple production workflows and task sequences are understood. 2.3 Teamwork and coordination in production processes are identified. 2.4 Workplace safety roles and responsibilities are understood and followed.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in jute processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance needs of tools and equipment are recognized.
4. Understand simple quality requirements in jute work	4.1 Basic quality standards in jute work are identified. 4.2 <u>Simple defects in jute products</u> are recognized. 4.3 Importance of maintaining quality during production is understood. 4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure

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	quality.
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Range of Variables

Variable	Range May include but not limited to:
1. Main activities	1.1 Cultivation and harvesting of jute 1.2 Retting and fibre extraction 1.3 Baling and grading 1.4 Spinning and yarn production 1.5 Weaving and fabric production 1.6 Dyeing, printing, and finishing 1.7 Packaging and export
2. Different stages of jute value chain	2.1 Raw jute production (farm level) 2.2 Fibre processing (retting, stripping, drying) 2.3 Baling and grading 2.4 Spinning into yarn 2.5 Weaving/knitting into fabric 2.6 Finishing and value addition 2.7 Distribution and export
3. Basic job roles in jute production	3.1 Raw jute handler 3.2 Machine operator 3.3 Quality checker/inspector 3.4 Dyeing and finishing worker 3.5 Packing and store assistant 3.6 Supervisor or line in-charge
4. Common tools used in jute processing	4.1 Scissors and cutting knives 4.2 Measuring tape and rulers 4.3 Hand hooks and tongs 4.4 Weighing scale 4.5 Hand brushes
5. Basic machines and equipment used in production	5.1 Carding machine 5.2 Drawing machine 5.3 Spinning machine 5.4 Looms (handloom/power loom) 5.5 Dyeing vats or tanks 5.6 Baling press machine
6. Simple defects in jute products	6.1 Uneven thickness or slub in yarn 6.2 Broken fibres or yarn ends 6.3 Colour variation or patchiness 6.4 Holes or tears in fabric 6.5 Mis weave or weaving faults

Curricular Content Guide

1. Underpinning Knowledge	1.1 Overview of the jute sector within national and global industry context. 1.2 Main activities and stages of the jute value chain from
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	raw material to finished products. 1.3 Basic job roles, responsibilities, and workflows in jute production processes. 1.4 Importance of teamwork, coordination, and workplace safety in production settings. 1.5 Identification and functions of common tools, machines, and equipment used in jute work. 1.6 Basic maintenance awareness for tools and equipment used in production. 1.7 Understanding of basic quality standards, product defects, and quality expectations. 1.8 Principles of good workmanship and maintaining quality during jute production processes.
2. Underpinning Skills	2.1 Understanding basic job roles and responsibilities in jute production processes 2.2 Following simple production workflows, task sequences, and teamwork practices 2.3 Applying workplace safety roles and responsibilities in production activities 2.4 Understanding basic functions and maintenance needs of tools and equipment 2.5 Applying quality awareness during production to meet expected standards 2.6 Checking work output against basic quality requirements 2.7 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified the jute sector and its main activities
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	1.2 understood basic roles in jute production processes 1.3 recognized common tools and equipment used 1.4 understood simple quality requirements in jute work
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN JUTE SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-DBP-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in Jute sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 Basic concepts of environmental sustainability and green practices are explained clearly. 1.2 <u>Environmental impacts of jute production</u> processes are identified. 1.3 <u>Resource efficiency concepts</u> are interpreted. 1.4 Waste reduction, reuse, and recycling principles are described in relation to jute operations. 1.5 <u>Eco-friendly practices in the jute sector</u> are recognized and related to workplace activities.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw materials are used efficiently according to workplace requirements. 2.2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2.3 Machines and equipment are operated in energy-efficient modes where applicable. 2.4 Material wastage is minimized through careful handling and accurate measurement. 2.5 <u>Resource-saving practices</u> are applied where possible. 2.6 Leakage, spillage, and wastage of materials are promptly controlled and reported. 2.7 <u>Workplace practices</u> are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in jute operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

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	3.6 Hazardous or chemical waste is handled carefully according to safety guidelines.
	3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables

Variable	Range (May include but not limited to):
1. Environmental impacts of jute production	1.1 Water pollution from retting and washing 1.2 Chemical discharge from dyeing and finishing 1.3 Solid waste generation (jute dust, fibre waste) 1.4 Air pollution from boilers and machines 1.5 Energy consumption impacts
2. Resource efficiency concepts	2.1 Water conservation techniques 2.2 Energy-efficient machine operation 2.3 Optimized chemical usage 2.4 Material loss reduction 2.5 Process efficiency improvement
3. Eco-friendly practices in the jute sector	3.1 Use of natural dyes or low-impact dyes 3.2 Biodegradable materials use 3.3 Compliance with environmental regulations 3.4 Proper effluent treatment practices 3.5 Energy-saving production methods
4. Unnecessary consumption of resources	4.1 Overuse of water or chemicals 4.2 Idle machine running 4.3 Excess raw material usage 4.4 Poor process planning or duplication of work
5. Resource-saving practices	5.1 Reuse of rinse water where applicable 5.2 Recycling of jute waste and scraps 5.3 Reusing packaging materials 5.4 Recovery of usable materials from waste
6. Workplace practices	6.1 Following environmental policies and SOPs 6.2 Proper waste segregation and disposal 6.3 Use of eco-friendly materials 6.4 Compliance with environmental regulations 6.5 Maintaining clean and safe work environment
7. Types of waste	7.1 Jute fibre waste and dust 7.2 Yarn and fabric off-cuts 7.3 Chemical waste (dyes, salts, auxiliaries) 7.4 Wastewater from processing 7.5 Packaging waste (poly bags, cartons) 7.6 Sludge from effluent treatment
8. Reusable and recyclable materials	8.1 Jute fibre scraps for reprocessing 8.2 Reusable packaging materials 8.3 Recyclable plastics and paper 8.4 Recoverable chemical containers (if applicable)

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Curricular Content Guide

1. Underpinning Knowledge	1.1 Concepts of environmental sustainability, green practices, and eco-friendly workplace behavior. 1.2 Environmental impacts of jute production and related industrial processes. 1.3 Principles of resource efficiency, including water, energy, and raw material conservation. 1.4 Waste reduction strategies such as reuse, recycling, and minimizing material wastage. 1.5 Efficient use of machines and equipment with focus on energy-saving practices. 1.6 Identification, segregation, and safe handling of different types of waste in jute operations. 1.7 Proper waste storage, disposal methods, and compliance with environmental regulations. 1.8 Hazardous waste handling and application of workplace environmental safety procedures.
2. Underpinning Skills	2.1 Applying resource efficiency principles in workplace operations 2.2 Applying waste reduction, reuse, and recycling practices in jute production activities 2.3 Promoting eco-friendly practices in daily workplace tasks 2.4 Using water, energy, and raw materials efficiently according to workplace requirements 2.5 Minimizing unnecessary consumption and wastage of resources during operations 2.6 Operating machines and equipment in energy-efficient modes where applicable 2.7 Handling materials carefully to reduce wastage and improve accuracy 2.8 Controlling leakage, spillage, and resource wastage and reporting issues promptly 2.9 Segregating different types of waste generated in jute operations 2.10 Handling hazardous waste according to safety guidelines and regulations
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource	The following resources must be provided:

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Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET MATERIALS AND PROCESSES	Nominal Duration: 20 Hrs.	Unit Code: SICIP-JUTE-JBP-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret materials and processes. It specifically includes the task of identifying jute fibres and materials, identifying dyes and chemical types, identifying tools and equipment, describing dyeing and bleaching processes and describing printing process.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify jute fibres and materials	1.1 <u>Types of jute fibres</u> are identified based on source and characteristics. 1.2 Physical properties of jute fibres are described. 1.3 Different grades and qualities of jute are recognized. 1.4 <u>Raw materials</u> used in jute processing are identified.
2. Identify dyes and chemical types	2.1 <u>Types of dyes</u> used in jute processing are identified. 2.2 <u>Chemical agents</u> used in bleaching, dyeing, and printing are recognized. 2.3 Properties and functions of dyes and chemicals are described. 2.4 Safety and handling requirements for dyes and chemicals are identified.
3. Identify tools and equipment	3.1 <u>Tools and equipment used in jute dyeing, bleaching, and printing</u> are identified. 3.2 Functions of each tool and equipment are described. 3.3 Equipment is classified according to process stages 3.4 Operational features and basic specifications are recognized. 3.5 Safety requirements for using tools and equipment are identified.
4. Describe dyeing and bleaching processes	4.1 Stages of jute dyeing and bleaching processes are identified. 4.2 <u>Procedures for dyeing and bleaching</u> are described step by step. 4.3 <u>Process parameters</u> are described.
5. Describe printing process	5.1 <u>Types of printing methods</u> used in jute processing are identified. 5.2 Steps of the printing process are described sequentially. 5.3 <u>Printing materials</u> are identified and explained.

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	5.4 <u>Process parameters</u> are described.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Types of jute fibres	1.1 White jute 1.2 Tossa jute 1.3 Mesta jute 1.4 Blended jute fibres 1.5 Sisal jute 1.6 Rami jute
2. Raw materials	2.1 Jute yarn 2.1.1 Twine yarn 2.1.2 Single ply yarn 2.1.3 Multiple ply yarn 2.2 Jute fabric 2.2.1 Hessian 2.2.2 Sacking 2.2.3 Canvas 2.3 Blended fabrics
3. Types of dyes	3.1 Direct dyes 3.2 Reactive dyes 3.3 Vat dyes 3.4 Pigment dyes 3.5 Azo-free dyes
4. Chemical agents	4.1 Bleaching agents 4.1.1 Hydrogen peroxide 4.1.2 Sodium hypochlorite 4.2 Mordants and fixing agents 4.3 Wetting agents and detergents 4.4 Softeners and auxiliaries
5. Tools and equipment used in jute dyeing, bleaching, and printing	5.1 Dyeing vats and tanks 5.2 Boilers and heating units 5.3 Padding mangle 5.4 Washing and drying machines 5.5 Printing tables and screens 5.6 Measuring instruments 5.6.1 Thermometer 5.6.2 pH meter 5.6.3 Weighing scale
6. Procedures for dyeing and bleaching	6.1 Pre-treatment 6.1.1 Scouring 6.1.2 Washing 6.2 Dye preparation and application 6.3 Bleaching process (oxidative/reductive) 6.4 Washing and rinsing 6.5 Drying and finishing

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7. Process parameters	7.1 Temperature 7.2 Time 7.3 pH level 7.4 Chemical concentration 7.5 Material-to-liquor ratio
8. Types of printing methods	8.1 Block printing 8.2 Screen printing 8.3 Roller printing
9. Printing materials	9.1 Printing pastes 9.2 Pigments and dyes 9.3 Binders and fixers 9.4 Thickeners 9.5 Water softener
10. Process parameters	10.1 Paste viscosity 10.2 Drying temperature 10.3 Curing conditions 10.4 Pressure and alignment 10.5 Fixation time

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types, sources, physical properties, and grading of jute fibres and raw materials. 1.2 Basic understanding of dyes, chemicals, and their functions in jute processing. 1.3 Safety handling requirements for dyes, chemicals, and related materials. 1.4 Identification and functions of tools and equipment used in dyeing, bleaching, and printing. 1.5 Classification of equipment based on process stages and operational features. 1.6 Knowledge of dyeing and bleaching processes, including key steps and procedures. 1.7 Understanding of jute printing methods, materials, and sequential process flow. 1.8 Awareness of process parameters and their importance in ensuring quality outcomes.
2. Underpinning Skills	2.1. Identifying types of jute fibres based on source, grade, and characteristics 2.2. Describing physical properties, quality grades, and raw materials of jute 2.3. Identifying dyes and chemical agents used in bleaching, dyeing, and printing processes 2.4. Identifying tools and equipment used in jute dyeing, bleaching, and printing operations 2.5. Describing functions, classifications, and basic specifications of equipment 2.6. Identifying types of printing methods used in jute

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	processing 2.7. Understanding process parameters and their role in ensuring quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Jute fibers 4.3 Raw materials 4.4 Tools and equipment's used in dyeing, bleaching nad printing

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified jute fibres and materials 1.2 identified dyes and chemical types 1.3 identified tools and equipment 1.4 described dyeing and bleaching processes 1.5 described printing process
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PREPARE MATERIALS AND CHEMICALS	Nominal Duration: 20 Hrs.	Unit Code: SICIP-JUTE-JBP-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to prepare materials and chemicals. It specifically includes the task of preparing jute materials for processing, cleaning and sorting raw materials, mixing chemical solutions safely and labelling and storing prepared solutions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare jute materials for processing	1.1 Jute fabric is selected as per job requirement. 1.2 Raw materials are inspected for quality and contamination. 1.3 Jute materials are selected according to processing requirements. 1.4 Materials are cut, softened, or conditioned as required. 1.5 Prepared materials are arranged for further processing.
2. Clean and sort raw materials	2.1 <u>Cleaning methods</u> are applied to remove unwanted substances. 2.2 Materials are sorted based on type, grade, and usability. 2.3 Defective or unsuitable materials are identified and separated. 2.4 Cleaned and sorted materials are stored appropriately.
3. Mix chemical solutions safely	3.1 Safety procedures and <u>personal protective equipment (PPE)</u> are applied during mixing. 3.2 <u>Required chemicals</u> are selected according to process specifications. 3.3 Measurements are taken accurately following <u>standard formulas</u> . 3.4 Chemicals are mixed in correct proportions and sequence. 3.5 Prepared solutions are checked for consistency and suitability.
4. Label and store prepared solutions	4.1 Prepared chemical solutions are labeled with required information. 4.2 Labels include details such as name, concentration, and date. 4.3 <u>Storage conditions</u> are maintained according to <u>safety guidelines</u> . 4.4 Containers are sealed and handled properly to prevent

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	contamination.
	4.5 Stored solutions are monitored for safety and usability.

Range of Variables

Variable	Range (Includes but not limited to):
1. Cleaning methods	1.1 Dry cleaning 1.2 Mechanical cleaning 1.3 Chemical cleaning 1.4 Scouring 1.5 Washing 1.6 Degumming
2. Personal protective equipment (PPE)	2.1 Gloves (chemical-resistant) 2.2 Safety goggles / face shield 2.3 Masks / respirators (for fumes and dust) 2.4 Aprons and protective clothing 2.5 Safety boots
3. Required chemicals	3.1 Sodium hydroxide (NaOH) 3.2 Sodium carbonate (Na ₂ CO ₃) 3.3 Sodium silicate (Na ₂ SiO ₃) 3.4 Detergent 3.5 Wetting agent 3.6 Bleaching powder 3.7 Hydrogen peroxide (H ₂ O ₂) 3.8 Acetic acid 3.9 Sodium hypochlorite 3.10 Sodium sulphite 3.11 Enzymes
4. Standard formulas	4.1 Dye-to-water ratio (e.g., material-to-liquor ratio) 4.2 Chemical concentration calculations (percentage solutions) 4.3 Mixing ratios for dyes, bleaching agents, and auxiliaries 4.4 Temperature and time parameters for processing
5. Storage conditions	5.1 Proper labeling of chemicals and solutions 5.2 Storage in cool, dry, and well-ventilated areas 5.3 Separation of incompatible chemicals 5.4 Use of sealed, corrosion-resistant containers 5.5 Shelf-life monitoring and stock rotation (First in First Out-FIFO method)
6. Safety guidelines	6.1 Wear personal protective equipment (PPE) 6.2 Follow machine operating instructions 6.3 Handle chemicals safely during mixing 6.4 Following Material Safety Data Sheet (MSDS) instructions 6.5 Ensure proper ventilation in the workplace 6.6 Use machine safety guards and devices

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	6.7 Store and label chemicals properly 6.8 Dispose of chemical waste safely 6.9 Maintain workplace hygiene and cleanliness 6.10 Follow emergency procedures (spill, fire, first aid) 6.11 Report hazards and accidents immediately
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of jute material selection, inspection, and preparation for processing. 1.2 Understanding of raw material quality, contamination types, and sorting criteria. 1.3 Methods of cleaning, grading, and safe storage of jute materials. 1.4 Basic processes of cutting, softening, and conditioning jute materials. 1.5 Knowledge of chemicals used in jute processing and safe handling practices. 1.6 Principles of measuring, mixing, and preparing chemical solutions in correct proportions. 1.7 Safety procedures including PPE use and risk prevention during chemical handling. 1.8 Labeling, storage requirements, and monitoring practices for prepared chemical solutions.
2. Underpinning Skills	2.1 Inspecting raw materials for quality, defects, and contamination 2.2 Preparing jute materials through cutting, softening, or conditioning as required 2.3 Sorting materials based on type, grade, and usability standards 2.4 Measuring chemicals accurately and mixing them in correct proportions 2.5 Preparing chemical solutions following proper sequence and standard practices 2.6 Checking consistency and suitability of prepared solutions 2.7 Storing and handling solutions safely to prevent contamination and hazards 2.8 Monitoring stored solutions to ensure safety and usability over time
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors

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	in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Personal protective equipment (PPE) 4.4 Jute fabric 4.5 Required chemicals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared jute materials for processing 1.2 cleaned and sorted raw materials 1.3 mixed chemical solutions safely 1.4 labelled and stored prepared solutions
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM BLEACHING OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-JUTE-JBP-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform bleaching operations. It specifically includes the task of preparing bleaching chemical solutions, loading materials into bleaching machine, carrying out bleaching operations and checking bleaching results and quality.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare bleaching chemical solutions	1.1 Personal protective equipment (PPE) is used in accordance with safety guidelines 1.2 <u>Appropriate bleaching agents and chemicals</u> are selected according to job requirements 1.3 Chemicals are measured accurately following <u>standard formulas</u> and specifications 1.4 Solutions are prepared using correct <u>mixing procedures and sequence</u> 1.5 <u>Required parameters</u> are maintained during preparation 1.6 Chemical solutions are labelled and handled following workplace procedures
2. Load materials into bleaching machines	2.1 Safety procedures are followed during loading operations 2.2 Jute materials are inspected for cleanliness and suitability before loading 2.3 <u>Materials loading procedures</u> are followed as per job requirement. 2.4 Machine loading capacity is followed to avoid overloading or damage 2.5 Materials are positioned according to machine requirements and process flow.
3. Carry out bleaching operations	3.1 <u>Machine settings</u> are adjusted according to process requirements and material type. 3.2 Bleaching machine is started and operated following standard operating procedures (SOPs). 3.3 <u>Process parameters</u> are maintained within required limits. 3.4 Adjustments are made as needed to ensure uniform and consistent bleaching results. 3.5 Machine performance is continuously monitored during the bleaching process.

	3.6 Waste, chemicals, and residues are handled and disposed of according to workplace procedures.
4. Check bleaching results and quality	4.1 Bleached materials are inspected for colour uniformity and required whiteness 4.2 Defects are identified and documented 4.3 Basic quality checks/tests are performed as per standards 4.4 Results are recorded accurately following workplace documentation procedures

Range of Variables

Variable	Range (Includes but not limited to):
1. Appropriate bleaching agents and chemicals	1.1 Hydrogen peroxide (H_2O_2) 1.2 Sodium hypochlorite ($NaOCl$) 1.3 Sodium silicate (Na_2SiO_3) 1.4 Sodium chlorite ($NaClO_2$) 1.5 Alkalis (caustic soda, soda ash) 1.6 Stabilizers, wetting agents, and sequestering agents
2. Standard formulas	2.1 Hydrogen peroxide (H_2O_2): 20–50 g/L 2.2 Sodium hydroxide ($NaOH$): 10–30 g/L 2.3 Sodium silicate: 10–20 g/L 2.4 Wetting agent: 5–10 g/L 2.5 Acetic acid (neutralization): 5–10 g/L 2.6 Bleaching liquor ratio: 1:10 to 1:20 (fibre : liquor) 2.7 H_2O_2 to water dilution: 1:20 to 1:30
3. Mixing procedures and sequence	3.1 Mixing (order) of chemical addition (Hot/cold water → auxiliaries → bleaching agent) 3.2 Controlled mixing to avoid rapid decomposition 3.3 Use of measured quantities and calibrated equipment 3.4 Safety precautions during chemical preparation
4. Required parameters	4.1 Temperature range (e.g., 80–100°C depending on process) 4.2 pH control (typically pH 10.5–11.5 (alkaline conditions for peroxide bleaching)) 4.3 Processing time 4.4 Water quality and hardness
5. Materials loading procedures	5.1 Proper loading capacity of machine (avoid overloading/underloading) 5.2 Uniform distribution of jute fibres/fabrics 5.3 Pre-wetting before bleaching 5.4 Ensuring free movement of materials inside machine
6. Machine settings	6.1 Temperature control settings 6.2 Timer and process duration settings 6.3 Agitation or rotation speed 6.4 Pressure settings (if applicable)

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7. Process parameters	7.1 Monitoring temperature, pH, and time during operation 7.2 Maintaining liquor ratio throughout the process 7.3 Observation of colour change and whiteness level 7.4 Adjustments during process (re-dosing, temperature correction)
8. Defects	8.1 Uneven bleaching / patchiness 8.2 Yellowing or fibre damage 8.3 Over-bleaching (loss of strength) 8.4 Residual stains or impurities

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of bleaching chemical selection, measurement, and safe solution preparation. 1.2 Knowledge of PPE use, safety procedures, and safe handling of bleaching chemicals. 1.3 Understanding of correct mixing procedures, process parameters, and solution consistency control. 1.4 Machine loading procedures, capacity limits, and safe handling of jute materials. 1.5 Operation of bleaching machines following SOPs and adjustment of process settings. 1.6 Monitoring and control of bleaching process parameters for uniform and effective results. 1.7 Waste, chemical residue handling, and environmental safety practices in bleaching operations. 1.8 Quality inspection methods, defect identification, and documentation of bleaching results.
2. Underpinning Skills	2.1. Measuring chemicals accurately according to standard formulas and specifications 2.2. Preparing bleaching solutions using correct procedures, sequence, and process parameters 2.3. Labeling and handling chemical solutions according to workplace safety procedures 2.4. Following correct loading procedures while maintaining machine capacity limits 2.5. Operating bleaching machines by setting and adjusting parameters as per SOPs 2.6. Handling and disposing of chemical waste and residues according to procedures 2.7. Inspecting bleached materials for color uniformity and required whiteness 2.8. Performing basic quality checks and recording results accurately as per standards
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

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	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure (samples) 4.3 Bleaching agents and chemicals 4.4 Bleaching machine

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared bleaching chemical solutions 1.2 loaded materials into bleaching machine 1.3 carried out bleaching operations 1.4 checked bleaching results and quality
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM DYEING OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-JUTE-JBP-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform dyeing operations. It specifically includes the task of preparing dye solutions and mixture, loading materials into dyeing machine, carrying out dyeing operations, checking colour quality and consistency.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare dye solutions and mixtures	1.1 <u>Dye types, chemicals, and auxiliaries</u> are selected according to production specifications. 1.2 Required quantities of dyes, water, and chemicals are measured accurately. 1.3 Dye solutions are prepared following <u>standard mixing procedures</u> . 1.4 Prepared dye mixtures are checked for uniformity before use.
2. Load materials into dyeing machine	2.1 <u>Dyeing machines</u> are cleaned before loading materials. 2.2 Materials are loaded evenly to ensure proper dye penetration. 2.3 Loading is done according to machine capacity and safety requirements. 2.4 Material positioning is verified to avoid entanglement and uneven dyeing.
3. Carry out dyeing operations	3.1 <u>Appropriate dyeing method</u> is selected based on material type and production requirements. 3.2 Dyeing machine is started and operated following standard operating procedures (SOPs). 3.3 <u>Machine parameters</u> are correctly adjusted. 3.4 Dye circulation and agitation are maintained throughout the dyeing process for uniform results. 3.5 Dye application is carried out evenly to ensure proper absorption and shade consistency. 3.6 Process conditions are monitored and adjusted to achieve required shade depth and fastness.
4. Check colour quality and consistency	4.1 Dyed materials are inspected for shade matching and uniformity. 4.2 Colour strength, brightness, and fastness are evaluated against standards. 4.3 Defects identified. 4.4 Required adjustments are recommended or applied if

	deviations occur.
	4.5 Quality inspection results are recorded according to workplace procedures.

Range of Variables

Variable	Range (Includes but not limited to):
1. Dye types, chemicals, and auxiliaries	1.1 Direct dyes 1.2 Reactive dyes 1.3 Vat dyes 1.4 Pigments 1.5 Salts (Glauber's salt, common salt) 1.6 Alkali (Soda ash, caustic soda) 1.7 Wetting agents 1.8 Dispersing agents 1.9 Leveling agents 1.10 Fixing agents
2. Standard mixing procedures	2.1 Pre-dissolving dye in water 2.2 Accurate sequential addition of chemicals 2.3 Continuous stirring or agitation 2.4 Maintaining required temperature 2.5 Ensuring complete dissolution (no lumps) 2.6 Filtration before use (if required) 2.7 Allowing solution to stabilize before dyeing
3. Dyeing machines	3.1 Dyeing Jigger Machine 3.2 Winch Dyeing Machine 3.3 Jet Dyeing Machine 3.4 Padder and Dryer Machine
4. Appropriate dyeing method	4.1 Exhaust dyeing method 4.2 Padding method 4.3 Continuous dyeing method 4.4 Batch dyeing process 4.5 Cold dyeing method 4.6 Hot dyeing method 4.7 Manual dyeing method (for small scale or sample production)
5. Machine parameters	5.1 Temperature 5.2 Pressure 5.3 Time

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of dye selection, chemical auxiliaries, and preparation of dye solutions. 1.2 Accurate measurement techniques for dyes, water, and chemicals in production processes. 1.3 Standard procedures for mixing dye solutions and
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	ensuring uniformity and stability. 1.4 Knowledge of dyeing machine operation, loading methods, and safety requirements. 1.5 Understanding of dyeing methods, process parameters, and their control for consistent results. 1.6 Importance of dye circulation, agitation, and even dye application for quality output. 1.7 Colour quality assessment including shade matching, fastness, brightness, and uniformity checks. 1.8 Documentation practices for recording dyeing results, defects, and corrective actions.
2. Underpinning Skills	2.1. Measuring dyes, water, and chemicals accurately for solution preparation 2.2. Preparing dye solutions using standard mixing procedures and techniques 2.3. Positioning materials properly to ensure uniform dye penetration and avoid entanglement 2.4. Operating dyeing machines following standard operating procedures (SOPs) 2.5. Adjusting machine parameters based on material type and production requirements 2.6. Monitoring and controlling process conditions to achieve required quality 2.7. Inspecting dyed materials for shade uniformity and color matching 2.8. Recording inspection results accurately as per workplace procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Dye types, chemicals, and auxiliaries 4.4 Dyeing machines

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared dye solutions and mixtures 1.2 loaded materials into dyeing machine
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	1.3 carried out dyeing operations 1.4 checked colour quality and consistency
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PRINTING OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-JUTE-JBP-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform printing operations. It specifically includes the task of preparing tools, materials and printing surface, preparing printing paste and colours, applying printing according to design, and maintaining pattern alignment and consistency.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare tools, materials, and printing surface	1.1 Printing tools, screens, and blocks are selected according to design requirements. 1.2 Jute material is checked for cleanliness and suitability for printing. 1.3 Printing surface is cleaned and prepared to ensure proper ink adhesion. 1.4 Tools and materials are arranged at the workstation for smooth operation.
2. Prepare printing paste and colours	2.1 <u>Printing paste ingredients</u> are selected correctly. 2.2 Required quantities are measured according to formulation standards. 2.3 Printing paste is mixed to achieve required consistency and viscosity. 2.4 Colour shades are prepared and matched with design specifications. 2.5 Prepared paste is checked for uniformity before application.
3. Apply printing according to design	3.1 <u>Appropriate printing method</u> is selected based on design and material type. 3.2 Printing paste is applied evenly according to design layout. 3.3 Colour application is carried out carefully to avoid bleeding. 3.4 Printing is performed following standard operating procedures.
4. Maintain pattern alignment and consistency	4.1 Design alignment is maintained to ensure accurate pattern registration. 4.2 Repeated patterns are checked for continuity and uniformity. 4.3 Printed output is inspected for <u>defects</u> . 4.4 Non-conforming prints are identified and separated from acceptable output.

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	4.5 Corrective actions are taken to maintain consistency and quality of printing.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Printing paste ingredients	1.1 Dyes 1.2 Pigments 1.3 Binders 1.4 Auxiliaries 1.5 Water softening agent
2. Appropriate printing method	2.1 Block printing 2.2 Screen printing 2.3 Roller printing
3. Defects	3.1 Uneven print shade 3.2 Blurring 3.3 Colour bleeding 3.4 Patchy printing 3.5 Missing print areas 3.6 Pinholes or spots 3.7 Streaks or lines 3.8 Overprinting or double impression 3.9 Poor colour fixation

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of selecting printing tools, screens, blocks, and materials based on design requirements. 1.2 Preparation of jute surfaces for printing, including cleaning and ensuring print readiness. 1.3 Basics of printing paste formulation, ingredient selection, and viscosity control. 1.4 Techniques for accurate measurement, mixing, and shade preparation of printing colours. 1.5 Understanding of different printing methods and their application on jute materials. 1.6 Principles of even colour application and prevention of defects such as bleeding or smudging. 1.7 Methods for maintaining design alignment, pattern continuity, and print consistency. 1.8 Quality inspection, defect identification, and corrective actions in printing operations.
2. Underpinning Skills	2.1 Selecting printing tools, screens, and blocks according to design requirements 2.2 Arranging tools and materials at the workstation for smooth workflow 2.3 Preparing printing paste using correct ingredients and

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	standard formulations 2.4 Measuring and mixing printing paste to achieve required consistency and viscosity 2.5 Applying printing paste evenly according to design layout 2.6 Maintaining design alignment and accurate pattern registration 2.7 Separating non-conforming prints from acceptable outputs 2.8 Applying corrective actions to maintain printing quality and consistency
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Printing tools, screens, and blocks 4.3 Printing paste ingredients

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared tools, materials, and printing surface 1.2 prepared printing paste and colours 1.3 applied printing according to design 1.4 maintained pattern alignment and consistency
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM FINISHING OPERATIONS	Nominal Duration: 20 Hrs.	Unit Code: SICIP-JUTE-JBP-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform finishing operation. It specifically includes the task of preparing for the works, identifying finishing process, carrying out finishing and cleaning and storing tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for the works	1.1 Personal protective equipment (PPE) is worn according to safety requirements. 1.2 Required <u>tools, equipment</u> , and materials are selected and prepared. 1.3 Jute materials are checked for cleanliness and readiness for finishing. 1.4 Work area is organized to ensure safe and efficient operation.
2. Identify finishing process	2.1 <u>Finishing process</u> is identified according to product specifications. 2.2 Process parameters are determined based on material type and desired finish. 2.3 Quality requirements and finishing standards are confirmed. 2.4 Safety and environmental requirements are reviewed before starting the process.
3. Carry out finishing	3.1 Finishing equipment is set and operated according to standard procedures. 3.2 Finishing process is carried out according to product specifications. 3.3 <u>Process conditions</u> are maintained within required limits. 3.4 Finished materials are inspected for appearance, texture, and quality.
4. Clean and store tools and equipment	4.1 Tools and equipment are cleaned after completion of finishing operations. 4.2 Residual chemicals and waste materials are disposed of according to workplace procedures. 4.3 Tools and equipment are inspected for damage or wear. 4.4 Faults or defects are reported to appropriate personnel. 4.5 Tools and equipment are stored in designated locations properly.

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Range of Variables

Variable	Range (Includes but not limited to):
1. Tools, equipment	1.1 Paper tube 1.2 Paper Cartoon 1.3 Danach 1.4 Polyethene 1.5 Marker pen 1.6 Cover 1.7 Scissor 1.8 Trolly 1.9 Digital Scale 1.10 Nylon tape 1.11 Measuring tape
2. Finishing process	2.1 Damping 2.2 Calendaring/ drying 2.3 Folding 2.4 Rolling 2.1 Packing
3. Process conditions	3.1 Temperature/Moisture 3.2 Pressure 3.3 Time

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of workplace safety, PPE use, and preparation for finishing operations. 1.2 Knowledge of tools, equipment, and materials used in jute finishing processes. 1.3 Understanding of finishing process types, parameters, and product specifications. 1.4 Basics of material preparation and quality requirements before finishing. 1.5 Operation of finishing equipment following standard procedures and safety guidelines. 1.6 Methods for monitoring process conditions and ensuring consistent finishing quality. 1.7 Inspection techniques for assessing appearance, texture, and overall product quality. 1.8 Procedures for cleaning, maintenance, storage of tools, and safe waste disposal.
2. Underpinning Skills	2.1 Checking jute materials for cleanliness and readiness before finishing processes 2.2 Organizing the work area to ensure safe and efficient operation 2.3 Confirming quality standards and finishing requirements before starting work 2.4 Carrying out finishing processes in line with product

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	specifications 2.5 Maintaining required process conditions for consistent finishing quality 2.6 Inspecting finished materials for appearance, texture, and overall quality 2.7 Inspecting tools and reporting faults or defects when identified 2.8 Storing tools and equipment properly in designated locations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Required tools, equipment 4.4 Finishing equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for the works 1.2 identified finishing process 1.3 carried out finishing 1.4 cleaned and stored tools and equipment
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Jute Bleaching, Dyeing, and Printing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	1
2.	Multimedia projector & screen	1
3.	Scanner	1
4.	Printer	1
5.	Jigger Machine	1
6.	Padder and Dryer Machine	1
7.	Electronic weighing scale	2
8.	pH meter	2
9.	Thermometer set	5
10.	Flatbed printing table	4
11.	Screen printing frames (assorted)	12
12.	Squeegee sets	12
13.	Ink/paste mixing machine	2
14.	Printing drying racks	2
15.	Drying chamber / drying room	1
16.	Hot air blower / industrial dryer	2
17.	Folding & inspection tables	3
18.	Fabric rolling machine	1
19.	Boiler (steam generation)	1
20.	Water pump system	1
21.	Chemical storage cabinets	2
22.	Fire extinguishers	2
23.	Fabric inspection light box	1
24.	Moisture meter	2
25.	GSM weighing scale	2

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

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The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Jute Dyeing, Bleaching and Printing** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.